

Evaluating RHC, Simulated Annealing, Genetic Algorithms, and MIMIC on Neural Network Weights

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I. EDA ON DATASETS

Exploratory data analysis (EDA) of the Adult income dataset shows that the dataset consists of 45,222 data points with 15 features. These are a mix of categorical variables (workclass, occupation, marital-status etc.) and continuous variables (age, capital-gain/capital-loss, hours-per-week, etc.). Some strong financial indicators like capital-gain and capital-loss, as in Figure 1, show an extreme right skew and sparsity with around 75% of the reported data being 0. Due to this sparsity, the model must often rely on other less reliable features such as age and education to predict income class. Many categorical variables have imbalanced frequencies, examples being white in race (86%), male in sex (2:1), and United States in native-country. Additionally, several categorical features contain many low-frequency categories, resulting in sparse high-dimensional representations after encoding. The target variable (income) also has a strong class imbalance with approximately a 3:1 ratio between $\leq 50K$ vs $> 50K$ income. Income patterns appear to arise from conditional interactions between features such as education, occupation, and age rather than linear relationships. Given these factors, pruned decision trees and SVMs are expected to perform the best as they can capture conditional structure and are robust to noise. In contrast, kNN is more sensitive to high-dimensional features and Neural Networks are most sensitive to class imbalance.

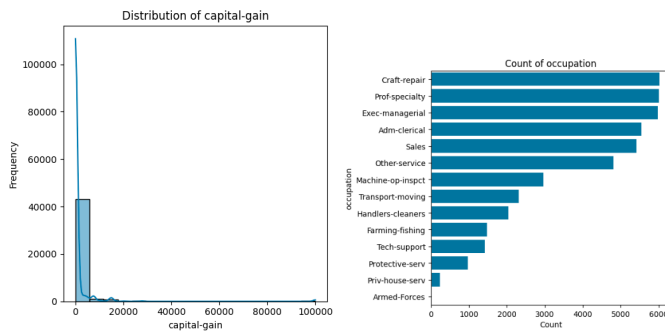


Fig. 1: Distribution of the capital-gain and occupation features in the Adult dataset.

EDA on the Wine Quality dataset shows us that the dataset has 6497 wines with 11 features. The class and quality features are identical and one of them will need to be dropped before training to avoid leakage. There are no missing values. Aside from type, all of the features are continuous variables. Several of the features are right-skewed. Some examples shown through diagrams are residual_sugar,

free_sulfur_dioxide, and total_sulfur_dioxide. The features free_sulfur_dioxide and total_sulfur_dioxide are strongly correlated as seen in the heat map in Figure 2. Alcohol is strongly negatively correlated with density, and there are moderate associations with quality as higher alcohol and sulphates tend to align with better quality. Higher volatile_acidity and type (1 - white wine, 0 - red wine) tend to align with lower quality. Multicollinearity between other features in the dataset is present but not extreme beyond these. Most feature-quality correlations are moderate between 0.3-0.4 and highly collinear groups are only weakly correlated with quality. There seems to be a large variance due to human perception, noise and unobserved variables. The class balance of the target variable (wine) is skewed towards mid-scored (3-5). Due to this, accuracy is less informative and F1 score will be more useful in assessing performance of the models. Overall, the features in the dataset provide little information to predict the target variable which may lead to the models not having very high performances.

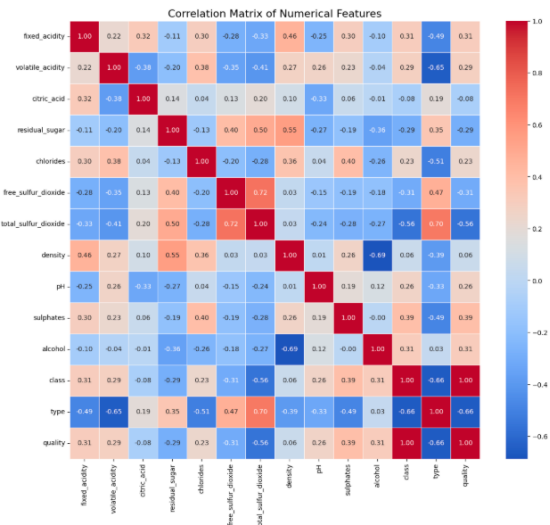


Fig. 2: Correlation matrix for Wine Quality dataset.

II. HYPOTHESES

The EDA on the Adult dataset shows us that the dataset is high-dimensional and has sparse features. Features like capital-gain and capital-loss are rare but influential and the target labels are imbalanced. Due to these properties, training is unstable as gradients are noisy and inconsistent. I hypothesize that optimizers, especially Adam, will improve

the macro-F1 and convergence time of the model as momentum, bias-correction, and flexible step-sizes will help smooth out the irregular gradients and improve convergence. However, random optimization on the last few layers of the network will have limited impact as the instability mainly originates earlier in the network due to the problem being ill-conditioned. Regularization will probably have a large impact on generalization by preventing the model from overfitting to the intricate patterns in the $\leq 50k$ class.

The EDA on the Wine dataset shows us that it is low-dimensional and mainly has continuous features. There are only moderate correlations between input data and quality label. The target is noisy and influenced by the subjective nature of wine tasting. The problem likely has many shallow local-minima caused by these weak feature-target correlations. Therefore, I hypothesize that applying randomized optimization to the final layers will help the network escape these shallow minima and improve generalization by promoting exploration that SGD may fail to achieve.

III. METHODS

On both the Adult Income and Wine Quality datasets, a deep neural network was first trained using stochastic gradient descent (SGD), particularly using Pytorch to do this. Data was split into 70% training, 15% validation, and 15% test sets using stratified sampling. For the Adult dataset, categorical features were one-hot encoded and numerical features were standardized, producing an input dimensionality of 103 after preprocessing. For the Wine dataset, all 11 input features were numerical and standardized directly. The NN used on both datasets consists of two hidden layers of sizes 150 and 100 with ReLU activations. In the Adult dataset this resulted in 30,902 parameters, the input-to-first layer contains 15,302 parameters, the first-to-second layer contains 15,600 parameters. In the Wine dataset, the network contained approximately 17,858 total parameters: 1,950 parameters are frozen from input to the first hidden layer and 15,908 parameters from the first to second hidden layer are trainable.

The three Randomized Optimization (RO) algorithms are then applied to the last two layers of the trained network separately. Three copies of the trained network are created on both datasets. All RO methods optimize the cross-entropy loss on the validation set with L2 regularization and each evaluation of this objective was counted as a function evaluation. Random Hill Climbing (RHC) used Gaussian weight perturbations with initial step size $\sigma_0 = 0.01$, which is geometrically decayed by $\gamma = 0.995$ to a minimum of 1×10^{-5} , with 5 independent restarts and a greedy acceptance rule. Simulated Annealing (SA) used Gaussian perturbations with initial $\sigma_0 = 0.05$ and a minimum of $\sigma = 1 \times 10^{-5}$, with temperature initialized at 5.0 and decayed using $T_t = T_0(0.999^t)$, allowing uphill movements via Metropolis criterion. Function evaluation budgets for RHC and SA were 2000 each.

The Genetic Algorithm (GA) maintained a population of 60 candidates initialized with Gaussian perturbations around

the baseline weights. Tournament selection ($k=3$) was used to choose parents, and offspring were generated using uniform interpolation crossover of the form $\lambda a + (1 - \lambda)b$, where λ was sampled independently per weight. Mutation was applied with probability 0.10 per parameter using Gaussian noise with an initial $\sigma = 1 \times 10^{-2}$ that decayed multiplicatively by 0.995 each generation to a minimum of 1×10^{-5} . Elitism preserved the top 2 individuals per generation. The total budget, including initialization, was limited to 3,000 function evaluations. The final models were evaluated on the test set.

In part 2 of the experiments, to investigate optimizer behavior, seven optimizers were applied to the baseline Pytorch neural network. These included stochastic gradient descent (SGD) without momentum ($lr = 1 \times 10^{-2}$, momentum=0.0), SGD with classical momentum ($lr = 1 \times 10^{-2}$, momentum=0.9), Nesterov accelerated gradient ($lr = 1 \times 10^{-2}$, momentum=0.9, nesterov=True), the Adam optimizer (lr and betas picked through tuning), a custom implementation of Adam with bias-correction terms removed, an Adam configuration with β_1 explicitly set to 0.0 to induce RMSProp-like behavior (lr and beta picked through tuning), and AdamW (lr and betas picked through tuning) employing decoupled weight decay. All optimizers were run for a maximum of 100 epochs unless terminated by the validation threshold condition. The threshold ℓ was defined once at the start of the study by using a baseline Adam run trained for 60 epochs at $lr = 1 \times 10^{-3}$ (seed=777) with a 5 epoch warmup and cosine annealing. Rather than the minimum along the trajectory, ℓ is set to 105% of the final validation objective. This is reused across all experiments. Cosine annealing and a 5 epoch warmup was also used across all experiments.

The validation objective is equal to the cross entropy loss. AdamW was run with its native decoupled weight decay (optimizer-side decay, not an L2 term in the loss), because this is the defining feature of AdamW. For the other optimizers (SGD, Momentum, Nesterov, Adam variants), optimizer weight_decay=0 and the same L2 penalty ($\lambda||\theta||^2$) was applied directly in training loss so they optimize an equivalent regularized objective. The optimizers were then evaluated with CE only, this ensured that differences in time-to- ℓ , convergence, and generalization reflected the optimizer behavior while still implementing AdamW. To assess parameter sensitivity in adaptive optimizers, grid-based sweeps were performed for Adam and AdamW. These sweeps evaluated learning rates α drawn from $\{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}, 1 \times 10^{-2}, 3 \times 10^{-2}\}$ against β_1 values $\{0.0, 0.5, 0.9, 0.95\}$ while fixing β_2 to 0.999, and separately against β_2 values $\{0.9, 0.99, 0.999, 0.9995\}$ while fixing β_1 to 0.9. Each configuration in the sensitivity analysis was trained for 30 epochs, and the best validation objective across those epochs was used to construct heatmaps. To examine stability, each of the optimizers SGD, Adam, and AdamW was additionally run across ten seeds (1-10) under identical settings for 80 epochs.

In part 3, a regularization study was conducted on the Adam-trained MLP using the best hyperparameters found in part 2. All experiments were trained with the Adam optimizer

using a batch size of 256, and a strictly enforced compute budget of 60 epochs. The objective was to isolate the effect of different regularization mechanisms while holding the architecture, optimizer settings, data splits, and training duration constant. Model selection was performed using validation macro-F1 from the best checkpoint within the 60-epoch budget. Five regularization strategies were evaluated independently via coarse grid search. First, L2 regularization was applied through Adam’s coupled weight decay parameter λ , modifying the loss to $\mathcal{L}(\theta) = \text{CE}(\theta) + \lambda \|\theta\|_2^2$, where θ denotes all trainable parameters. The grid explored was $\lambda \in \{0, 10^{-5}, 5 \times 10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$. These magnitudes are known to meaningfully constrain parameter growth in MLPs without suppressing rare features. Second, dropout was applied after each hidden ReLU activation with dropout probability p such that each hidden unit was zeroed independently with probability p during training. No dropout was applied to input and output layers. The grid evaluated was $p \in \{0.0, 0.1, 0.2, 0.3, 0.5\}$. This range of values is light to fairly strong while avoiding extreme values known to harm models. Third, label smoothing was incorporated into the cross-entropy loss. Smoothed targets were defined as $\tilde{y}_k = (1 - \varepsilon)\mathbf{1}_{k=y} + \frac{\varepsilon}{K}$, where ε is the smoothing parameter. The grid explored was $\varepsilon \in \{0.0, 0.05, 0.1, 0.2\}$. This range was chosen as Adult shows class imbalance making moderate smoothing appropriate for reducing overconfidence without erasing minority class. Large values were avoided as they would flatten distribution aggressively. Fourth, input Gaussian noise was added during training only, generating perturbed inputs $x_{\text{noisy}} = x + \mathcal{N}(0, \sigma^2)$, where σ denotes the noise standard deviation. The grid evaluated was $\sigma \in \{0.0, 0.01, 0.02, 0.05\}$. These magnitudes inject small perturbations while leaving one-hot dimensions unaltered. Fifth, early stopping was examined using patience parameter τ , with $\tau \in \{\text{None}, 3, 5, 10\}$. These were chosen as Adult dataset has a tendency to peak in validation macro-F1 within the first 10 epochs.

To tune the combination of regularization, greedy coordinate search was used that optimizes one type of regularization at a time while holding others fixed. Each grid was swept in turn starting with a fixed early stopping patience of 3 epochs. The value that maximized the validation macro-F1 was picked. At the end, the best final combination was selected. This was repeated a second time with no early stopping to ensure that a fair chance was given to the regularizers on a full budget run. The best final combination was again selected here by maximum macro-F1. The hardware class used for all experiments was CPU: Intel(R) Xeon(R) CPU @ 2.20GHz, OS Class: Ubuntu 22.04.5 LTS.

A dummy classifier was trained on both datasets as a sanity check. For the Wine quality dataset it resulted in 0.35 accuracy and an extremely low macro F1 of 0.064. This confirms that no leaky features were used and that the train/test split was performed prior to modeling. Similarly, for the Adult dataset, it had 0.75 accuracy but a macro F1 of 0.43 and zero recall for the >50K class.

IV. RANDOMIZED OPTIMIZATION

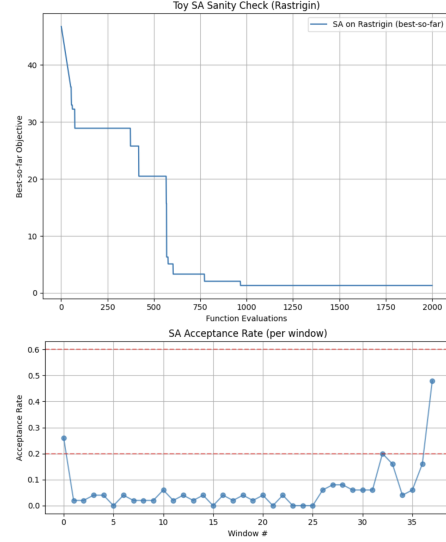


Fig. 3: Results of SA on toy Rastrigin experiment.

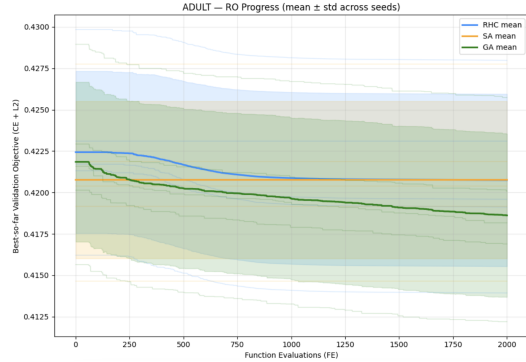


Fig. 4: RO progress curves against function evaluations on the Adult dataset. Settings: [RHC] restarts: 5 | $\sigma_{\min}$: $1 \cdot 10^{-5}$ | FE: 2000 [SA] T0: 5.0 | alpha: 0.999 | $\sigma_{\min}$: $1 \cdot 10^{-5}$ | FE: 2000 [GA] pop: 60 | tourn_k: 3 | pmut: 0.1 | σ_{mut} : 0.01 | σ_{decay} : 0.995 | $\sigma_{\min}$: $1 \cdot 10^{-5}$ | elites: 2 | FE: 2000

To ensure correct implementation of the SA algorithm, a toy SA was run on the two dimensional Rastrigin function as a sanity check. It showed successful behavior, the objective value is steadily lowered over the FE budget, moving from large exploratory steps at high temperature to smaller steps at lower temperature that exploit the minima. The best-so-far curve also converges to the global optimum. Therefore, the results of SA on the two datasets will reflect the optimization surface of the baseline solution rather than an issue with SA implementation.

On the Adult dataset, all three RO methods showed consistent behavior across the five seeds. This indicates that the parameters in the last two layers are already close to a stable local optimum that was found by the SGD baseline. This must have left little room for further improvements.

TABLE I: Performance Summary of RO Methods on Adult Dataset

Method	Best Val Loss	f1-macro	#Func Evals	Updates
Baseline	0.4208 ± 0.0048	0.7775 ± 0.0088	2000 ± 0	17 ± 0
RHC	0.4207 ± 0.0052	0.7801 ± 0.0076	1851 ± 333	1849 ± 335
SA	0.4208 ± 0.0048	0.7775 ± 0.0088	2000 ± 0	1999 ± 0
GA	0.4186 ± 0.0049	0.7804 ± 0.0077	2000 ± 0	1941 ± 0

The results were consistent with the hypothesis with limited impact seen. RHC (Random Hill Climbing) performs slightly but consistently better than the baseline. It improves test macro-F1 to 0.7801 ± 0.0076 . Since RHC has greedy acceptance, it adopts a solution only if it improves the objective. On this dataset, it has quick, early improvements as seen in the graph. The algorithm captures small downhill directions, followed by a plateau once all these are exhausted. The curve decreases and then stabilizes similarly. This reflects that the surroundings of the SGD solution have minor descending directions with little to exploit overall. The performance is stable with $\text{IQR} = 0.0034$.

The Genetic Algorithm (GA) has the strongest improvement and lowest validation objective curves with test macro-F1 0.7804 ± 0.0077 . The GA curve has staircase-like movements because each generation evaluates many candidates ($\text{pop}=60$) and applies variation operators (tournament selection, uniform crossover, and Gaussian mutation) that allow it to explore many directions in parameter space. This allows many solutions to explore the space in parallel and combine multiple small improvements. This is advantageous as it is difficult for a single trajectory to find a beneficial direction on the Adult dataset.

Simultaneous Annealing behaves very differently. After testing across a few Temperatures and increasing T to 5.0, slowing the temperature decay and increasing size of perturbations, the validation-FE curve remains flat. It produces mean macro-F1 values identical to the SGD baseline for all seeds (mean 0.7775 ± 0.0088). SGD already seems to have found a near-optimal solution. The lack of improvement does not indicate faulty implementation. The acceptance rates follow a good pattern with early windows showing high acceptance (0.95) allowing broad exploration and this drops through the runs to below 0.2 as the temperature cools. Even though SA accepts worse moves, the changes are too small to lead towards better regions. SA explores locally, but does not discover meaningful improvement directions. Unlike RHC, which keeps small gains, or GA, which explores multiple solutions in parallel, SA follows a single random path and ends up wandering near the same solution without lowering the best-so-far objective, leading to a flat curve.

Overall, all three methods show high stability across seeds with IQR values around 0.007-0.008 for macro-F1. This shows the dataset’s consistency across the stratified splits. Since only small performance improvements are made by RHC and SA, this shows that there is a narrow margin for improvement in the last two layers.

On the Wine dataset, the algorithms behave differently. The optimization surface seems to be more rugged and

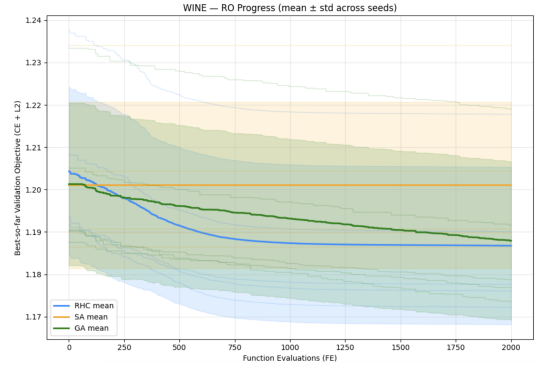


Fig. 5: RO progress curves against function evaluations on the Wine dataset. Settings: [RHC] restarts: 5 | $\sigma_{\min}$: 0.01 | γ : 0.995 | $\sigma_{\min}$: $1 \cdot 10^{-5}$ | FE: 2000 [SA] T0: 5.0 | α : 0.999 | $\sigma_{\min}$: 0.05 | $\sigma_{\min}$: $1 \cdot 10^{-5}$ | FE: 2000 [GA] pop: 60 | tourn_k: 3 | pmut: 0.1 | σ_{mut} : 0.01 | σ_{decay} : 0.995 | $\sigma_{\min}$: $1 \cdot 10^{-5}$ | elites: 2 | FE: 2000

TABLE II: Performance Summary of RO Methods on Wine dataset

Method	Best Val Loss	f1-macro	#Func Evals	Updates
Baseline	1.2010 ± 0.0196	0.2901 ± 0.0131	2000 ± 0	112 ± 0
RHC	1.1868 ± 0.0186	0.2899 ± 0.0189	2000 ± 0	1999 ± 0
SA	1.2010 ± 0.0196	0.2901 ± 0.0131	2000 ± 0	1999 ± 0
GA	1.1879 ± 0.0187	0.2901 ± 0.0134	2000 ± 0	1941 ± 0

inconsistent. Unlike hypothesized, none of the algorithms consistently improve the macro-F1 score. The dataset has 8 classes and more label noise which must result in a less smooth surface. None of the algorithms consistently outperform the SGD baseline, RHC and GA sometimes find small improvements against the baseline on some seeds. From the curve, we can see that RHC has gains at the start with a downward descent before hitting a plateau. On some seeds it slightly beats the SGD baseline (e.g., 0.3162 vs. 0.2963 baseline in seed 33) and in others it slips below the baseline (seed 22). This suggests that there are small improvements in the last two layers but they are sparse and irregular. The mean RHC macro-F1 (0.2899 ± 0.0189) is slightly less than baseline (0.2901 ± 0.0131) but the IQR is more unstable than Adult. This shows that the algorithm’s success is dependent on the seed rather than consistent global structure.

GA, although having performed best on Adult, is more inconsistent on Wine. GA’s curves show slow reductions in the validation objective, but these do not translate to a much higher macro-F1. The average macro-F1 (0.2901 ± 0.00134) is only exactly equal baseline but with slightly higher variance. This probably means that GA is better than RHC at exploring multiple regions simultaneously but the last two layers have few improvements that can be exploited. There seems to be more noise and uncertainty in the multiclass prediction especially due to the high class imbalance. The GA curve is above the RHC curve on this dataset unlike in the Adult showing that it performs worse than it overall.

SA behaves exactly like in Adult, showing a flat curve across all seeds and delivering the exact same results as the SGD baseline (0.2901 ± 0.00131). Initially, although SA accepts a lot of moves initially, the moves are not beneficial. They are neutral or uphill steps. The SGD solution seems to already sit near an optimal solution and the random perturbations never land on a downward trajectory that reduce loss. The algorithm starts by exploring broadly but is unproductive and then the temperature becomes too cold to escape the initial solution. This is why the results are identical to the SGD baseline. Overall, the stability across seeds shows that finetuning on Wine dataset is challenging. This partially contradicts the hypothesis that RO would more consistently outperform SGD on Wine dataset. Although the dataset likely contains shallow local minima due to weak correlations, the level of label noise limits the benefit of exploration of the optimization surface. RHC and GA occasionally discover slightly better solutions but these are dependent on the initialization seed. This suggests that local improvements may correspond to fitting noise. The improvements are not reliable unlike in Adult and the problem is volatile. IQRs are wider and mean differences are smaller between algorithms. GA, which performs well on Adult, has inconsistent gains. All three algorithms show weaker and less stable improvement than Adult.

V. ADAM/ADAMW HEATMAPS

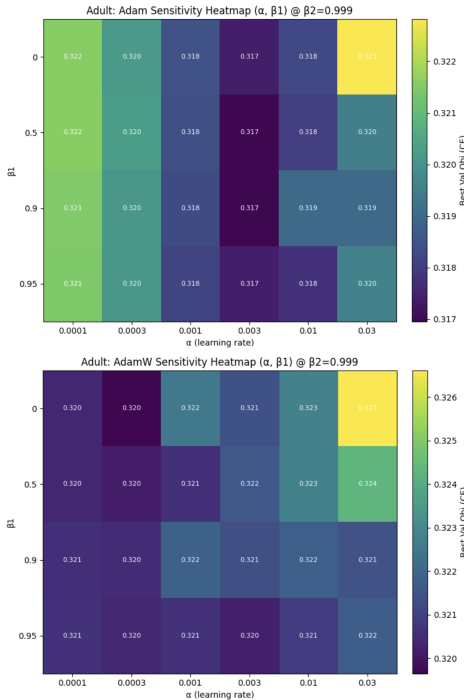


Fig. 6: Adam and AdamW sensitivity heatmaps on the Adult dataset.

The tuning heatmaps show that Adam performs well across set of learning rates and β_1 values with validation CE staying tightly clustered around 0.317-0.320 except where

learning rate becomes extremely large. Adam is able to take larger steps because its L2 regularization is coupled to the gradients. Weight shrinkage only happens in directions with supporting gradient direction. This makes higher β_1 values (0.9-0.95) stable, and allows α as high as 0.01 without destabilizing the update. The final configuration is $\alpha=0.01$, $\beta_1=0.95$, $\beta_2=0.999$.

AdamW heatmap is very different. The best region is at very small learning rates ($\alpha = 3 \cdot 10^{-4}$) and zero momentum $\beta_1=0$. This sensitivity is due to AdamW’s decoupled weight decay, the shrinkage term acts directly on the weights at every step regardless of gradient direction. This behavior is consistent with the formulation of AdamW where decoupled weight decay can amplify sensitivity to learning rate and momentum parameters in practice [1]. This was seen on an initial experiment where $\beta_1 > 0$ AdamW explodes after very quickly converging to ℓ . On the Adult dataset, where weights rely on small magnitudes, this can overpower the useful gradient signal when learning rate or momentum is high. As a result AdamW is very unstable at high β_1 , and high α and only works when $\beta_1=0$ and learning rate is low. The final best configuration is $\alpha=0.0003$, $\beta_1=0$, $\beta_2=0.9$. Adam is more naturally compatible with the Adult dataset, as it benefits from aggressive learning rates and strong momentum while AdamW needs very conservative settings.

VI. ADAM ABLATIONS RESULTS

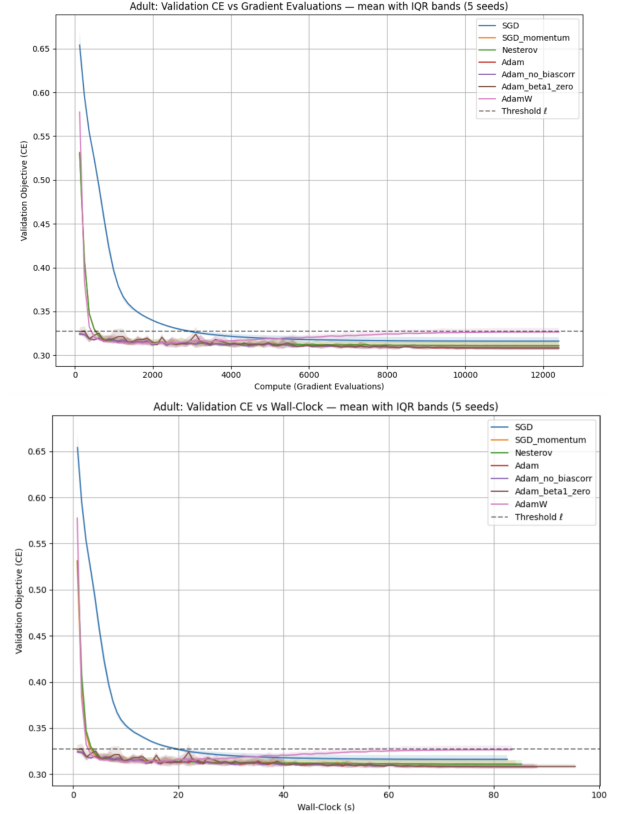


Fig. 7: Loss vs compute/wall-clock curves for optimizer experiments on Adult dataset across seeds.

TABLE III: Adult Part 2: Optimization Efficiency, Computational Budget, and Classification Metrics (Mean / Median / IQR)

Optimizer	Time to ℓ (s)	Steps to ℓ	Grad Evals	Best Val CE	Gen Gap (CE)	Test Macro-F1
Adam	0.98 / 0.82 / 0.02	148.8 / 124 / 0	12400 / 12400 / 0	0.308 / 0.309 / 0.005	0.010 / 0.013 / 0.009	0.79811 / 0.79527 / 0.00979
AdamW	3.51 / 3.89 / 0.97	496.0 / 496 / 0	12400 / 12400 / 0	0.313 / 0.313 / 0.005	0.053 / 0.052 / 0.008	0.79297 / 0.79450 / 0.00424
Adam ($\beta_1 = 0$)	1.15 / 0.83 / 0.80	173.6 / 124 / 124	12400 / 12400 / 0	0.308 / 0.309 / 0.004	0.010 / 0.012 / 0.007	0.79891 / 0.79784 / 0.00817
Adam (no bias)	1.05 / 1.05 / 0.35	148.8 / 124 / 0	12400 / 12400 / 0	0.308 / 0.308 / 0.005	0.010 / 0.011 / 0.008	0.79892 / 0.79503 / 0.01116
Nesterov	4.21 / 4.08 / 0.42	620.0 / 620 / 0	12400 / 12400 / 0	0.311 / 0.311 / 0.007	0.010 / 0.011 / 0.008	0.79782 / 0.79533 / 0.01102
SGD	21.41 / 20.32 / 3.09	3199.2 / 2976 / 372	12400 / 12400 / 0	0.316 / 0.316 / 0.007	0.001 / 0.001 / 0.008	0.79400 / 0.79177 / 0.00834
SGD (momentum)	4.01 / 4.11 / 0.51	620.0 / 620 / 0	12400 / 12400 / 0	0.311 / 0.311 / 0.007	0.010 / 0.011 / 0.008	0.79769 / 0.79510 / 0.00838

The experiment on the seven optimizers shows the effects that momentum, adaptive scaling, bias correction have on speed-to-convergence, stability and generalization of the Adult dataset. The speed to the set validation loss threshold ℓ has varying results. Vanilla SGD is the slowest, with a mean of 3199 steps. With momentum, this greatly reduces to 620 steps for both SGD+momentum and Nestrov, but momentum alone leads to a 5x speedup. In adaptive optimizers, there is another big jump, Adam (148 steps), Adam-no-biascorr (148 steps), and Adam with $\beta_1 = 0$ (178 steps) on average. There is a close to 5x speedup again with adaptive scaling. Adam-style optimizers seem to very efficient. Lack of bias correction does not affect the speed of the Adam optimizer. Both Adam and Adam-no-biascorr converge with identical steps to ℓ . However, AdamW takes longer with a mean of 496 steps. This is because decoupled weight decay shrinks the weights at every step which dampens the updates. Aside from Adam_beta1_zero (124) and SGD (372), IQR is 0 for all other optimizers indicating good consistency in convergence across seeds. This means that most optimizers reach the threshold in exactly the same number of steps, indicating that the early optimizer trajectory is highly deterministic once adaptive scaling is applied. Momentum methods also show relatively small dispersion in steps to ℓ compared to vanilla SGD. SGD’s larger IQR reflects sensitivity to initialization and noisier gradient updates.

Generalization behavior is measured by the gap between final train CE and final validation CE. Adam has a small gap of 0.0104. Adam-no-biascorr and Adam $\beta_1 = 0$ produce similarly small gaps (0.0105–0.0103). The momentum methods show similar small gaps (Nestrov/SGD+momentum - 0.0099), this is because they usually land in broader valleys. SGD has the smallest generalization gap of 0.00125, this is because its updates are small, noisy and conservative. This prevents it from overfitting the training distribution. AdamW shows the largest generalization gap (0.052) but a surprisingly tight IQR (0.008) in comparison, this means that AdamW is consistently biased. AdamW over-shrinks weights and over-regularizes the model due to decoupled weight decay, this leads to poor generalization. All the IQRs fall within a narrow band 0.0070-0.0083. This indicates good stability in behavior across seeds. We can also see this in the graph where the IQR band on the curves is very small to almost negligible. This suggests that the balance between training fit and validation performance is determined by

optimizer rather than random initialization.

Macro-F1 across seeds at the budget checkpoint on test fall within a narrow band. The Adam family consistently has the highest macro-F1 values (around 0.798–0.799). Adam’s adaptive per-coordinate scaling helps the network preserve the minority class recall. This is visible in Adam’s mean macro-F1 of 0.79811, Adam-no-biascorr at 0.79892, and Adam $\beta_1 = 0$ at 0.79891, all tightly clusters with moderate IQRs (0.008-0.011). The strong macro-f1 aligns with their fast convergence to ℓ and small generalization gaps from above as well. In contrast, momentum methods (SGD_momentum and Nestrov) have a slightly lower macro-F1 (0.79782 and 0.79400 respectively). This shows that momentum improves progress but Adam’s fine grained adaptivity works better in the imbalanced Adult dimensions. Vanilla SGD has a lower macro-F1 of 0.7940, this is consistent with its slower convergence and noisier trajectory. This prevents it from forming an equally strong decision boundary to separate the minority class with the same computational budget. AdamW has the most distinctive behavior, it’s macro-F1 of 0.7929 consistently underperforms Adam despite being extremely stable across seeds (IQR = 0.004). This indicates that AdamW’s solutions are consistently over-regularized. This follows from the effects of decoupled weight decay discussed earlier. This suppresses logit magnitude and reduces confidence which harms the minority class. The rest of the optimizers have slightly higher IQR values between 0.008-0.011 indicating that predictive performance is stable across seeds and the results are a good indication of the optimizer’s influence.

Overall, the behavior across the seven optimizers shows us that starting from SGD baseline, momentum has a great benefit on improving it. The steps-to- ℓ are reduced by 5x and there is a decent increase in macro-F1 from 0.7940 to 0.7977 (highest observed macro-f1 is 0.79892). Therefore, the bulk of the improvement from the baseline can be credited to momentum only. Adaptive scaling in the Adam family makes the optimizers both faster and more stable across seeds. All Adam variants reach ℓ in 124 steps in almost every seed (IQR = 0 for Adam and Adam-no-biascorr) which is 25x faster than SGD. Adam’s adaptivity ensures that rare but informative features receive scaled updates increasing F1-macro to be the highest across optimizers as discussed as well.

Bias correction does not matter in this task. Adam and

Adam-no-biascorr have the same steps-to- ℓ and IQR = 0, this shows that removing bias correction does not slow or destabilize the early-phase trajectory. Their generalization gaps are also nearly identical. The loss landscape is low-curvature and smooth so the stabilizing effect of bias-corrected momentum is not required. Decoupled weight decay behaves very differently from standard Adam in generalization. AdamW produces the worst generalization gap (0.0526) and the lowest macro-F1 among the adaptive methods, despite high stability. Decoupled weight decay interacts poorly with this dataset, resulting in an over-regularized solution. The macro-F1 results show that adaptivity is the main ingredient that improves both performance and robustness, momentum provides the first big jump after SGD.

These findings support the hypothesis made from EDA of the Adult dataset. The class imbalance and sparsity of the dataset do produce noisy and uneven gradients. This makes simple gradient descent inefficient. As hypothesized, optimizers that incorporate momentum and adaptive per-parameter scaling perform much better than vanilla SGD in both convergence speed and macro-F1. Momentum alone provides a major improvement by smoothing noisy updates and accelerating progress along consistent gradient directions. Adaptive optimizers such as Adam then build on this by rescaling updates per parameter, allowing the model to learn from rare but informative features like capital-gain and capital-loss.

VII. REGULARIZATION STUDY RESULTS

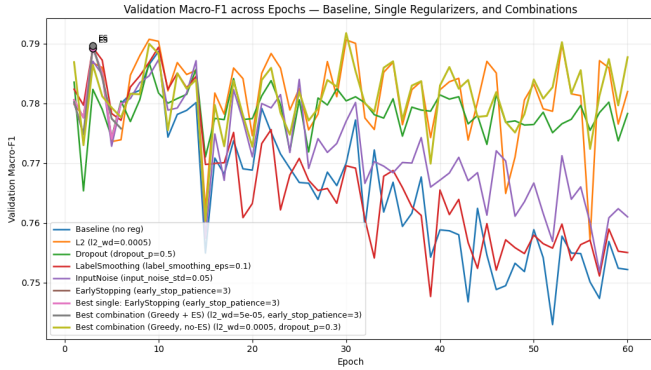


Fig. 8: Validation macro-F1 against curves epochs for all regularizers on Adult Dataset

TABLE IV: Comparison of Regularization Methods

Method	Best Val Loss	macro-f1	#Grad Evals	#Func Evals
Baseline (no reg)	0.320678	0.761080	7440	123
L2	0.317689	0.796800	7440	123
Dropout	0.318655	0.791418	7440	123
LabelSmoothing	0.415294	0.767090	7440	123
InputNoise	0.320261	0.774840	7440	123
EarlyStopping	0.320678	0.799699	744	15
Best single: EarlyStopping	0.320678	0.799699	744	15
Best combination (Greedy + ES)	0.320191	0.799004	744	15

Across all experiments on the Adult income dataset, there is a consistent pattern. The network reaches its best generalization performance very early in training and most

regularization techniques work by counteracting the strong overfitting that occurs after a few epochs. The baseline model, trained for the full 60 epochs without regularization has a test macro-F1 of 0.7611. The validation macro-F1 curve peaked around epoch 3 and then declined. This reflects that the Neural Network quickly memorizes the intricate patterns in the Adult dataset especially due to the the aggressive learning rate (0.003) and low $\beta_1 = 0.5$ found through tuning the Adam model. Both of these accelerate learning early on. The results support the hypothesis that regularization would improve macro-F1 as the model tends to overfit patterns favoring the majority class in the Adult dataset. Introducing L2 weight decay through Adam’s coupled weight decay parameter improved performance smoothly up to $5 \cdot 10^{-4}$ with a corresponding macro-F1 of 0.7986. This is because L2 regularization reduces the magnitude of weights, this prevents the model from fitting high variance patterns in the data. When the parameter becomes too large, the model underfits slightly, reducing macro-F1 to 0.7950.

Dropout has a beneficial effect as well, but the highest value (0.7916) was less than L2 regularization’s. The macro-F1 increased upto a value of $p = 0.2$ and drops afterwards. This means that higher dropout values slows learning too much when the model still runs up until 60 epochs. Label smoothing had a limited impact. Smoothing produced values around 0.76-0.77 which are close to the baseline value. This is expected as label smoothing mainly helps when labels are noisy or classes are ambiguous but adult is a clean binary classification task. Softening targets barely changes the decision boundary.

Gaussian noise also only provided small gains with highest being 0.7748 at $\sigma = 0.05$. This shows that small perturbations may slightly improve the model but large noise might drown out the already sparse signals in the Adult dataset. The strongest single method was early stopping which achieved a macro-F1 of 0.7997 with a patience of 3. This is a massive improvement over the baseline compared to other methods. This is consistent with what’s seen during training, validation macro-F1 is highest earlier in the graph (epoch 3-4), after which the model’s representation highly overfits the training data. Early stopping freezes the model when its generalization is best. This aligns with prior work showing that early stopping acts as an implicit regularizer by stopping training before the network begins fitting noise in the training data, thereby improving generalization [2].

The combination experiments show a consistent pattern of early stopping determining the final performance. When early stopping is enabled, other regularizers provide little additional benefit. In greedy search with early stopping, the procedure converges to the same configuration multiple times, early stopping = 3, a small L2 term ($5 \cdot 10^{-5}$) and no dropout, label smoothing, or noise (giving a macro-F1 of 0.7990). The dominant factor improving the generalization is the early peak in validation score around epoch 3. When combinations of regularizers are tested with no early stopping, the greedy search selects more heavy explicit regularization (L2 = $5 \cdot 10^{-4}$, dropout = 0.3) to counteract

overfitting. The final solution gives a good result (0.7969) but is slightly weaker than the solution with early stopping (0.7990). Label smoothing and input noise fail to enter optimal combinations because of reasons discussed above. Overall, overfitting happens rapidly and sharply on this dataset, making early stopping most effective. The final best combination of regularizers is L2 regularization = $5 \cdot 10^{-4}$, dropout= 0, label smoothing= 0, gaussian noise = 0, early stopping patience= 3.

VIII. FINAL MODEL WITH BEST TECHNIQUES

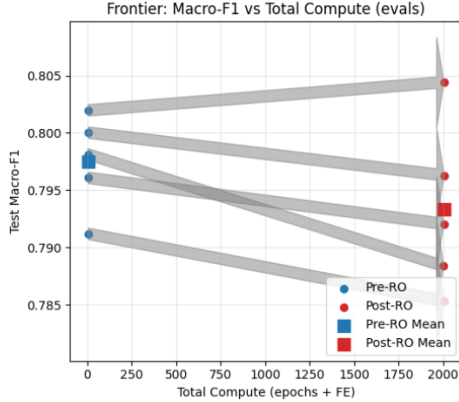


Fig. 9: Macro-F1 vs Total Compute Frontier graph for final model on Adult dataset.

Picking the best methods and hyperparameters from every part gives us the final model with Adam_no_biaccorr, EarlyStopping(p=3), and Genetic Algorithm (GA) Random Optimization. The Adam_no_biaccorr + EarlyStopping(p=3) model was first tested across five seeds, giving a mean macro-F1 of 0.7972 ± 0.0040 and accuracy 0.8499 ± 0.0025 . Applying GA-based RO to the last two layers (FE=2000) raises accuracy slightly 0.8533 ± 0.0023 (+0.0034) but reduces macro-F1 to 0.7936 ± 0.0076 (-0.0036) and increases macro-F1 variance (std 0.0040 to 0.0076). The compute tradeoff is very high, average evaluations increase from 5.2 epochs to 2005 (385x more), and wall-clock rises from 5.7s to 36.3s (6.3x slower), for at best a negligible accuracy gain and a small macro-F1 decline. Per-seed results show no consistent macro-F1 improvements (only seed 43 improved slightly from 0.8020 to 0.8044, while others fell). Given the Adult dataset’s imbalance and sparse but influential features, macro-F1 is the more informative metric; the post-RO drop and higher variance indicate no robustness or generalization benefit from last-layer-only RO, consistent with the idea that instability originates earlier in the network. The hypothesis on the Adult dataset is accepted. Adam_no_biaccorr+EarlyStopping(p=3) converges very quickly and increases macro-F1 compared to SL baseline performance (0.7901). RO clearly has little to no impact as hypothesized.

IX. CONCLUSION

The outcomes across the experiments reveal the characteristics of the datasets. They largely support the original

hypothesis. The Adult dataset is high dimensional, sparse and imbalanced. It generates noisy and inconsistent gradients, making optimization challenging and sensitive to adaptive mechanisms. As predicted, Adam and its variants consistently outperformed SGD and momentum-based methods in both convergence speed and macro-F1 score, because their per-parameter scaling and momentum stabilize the irregular gradients that arise from rare but influential one-hot-encoded features. Optimizers increase F1-score by +0.004-0.005 and convergence speed by upto 5x. Regularization, particularly early stopping and L2 weight decay greatly helped generalization (increasing macro-F1 by 0.036-0.039) on the model as it overfits very quickly. This is due to the aggressive learning rate and the dataset’s intricate patterns.

Randomized optimization (increases macro-F1 by +0.002-0.003 on average) on Adult behaved as hypothesized as well, its impact was limited. RHC and GA found only small, early improvements and quickly plateaued while SA remained flat. This indicates that after SGD training, the last two layers already lie in a narrow local basin with little exploitable downhill curvature. The instability of the problem mainly comes from the earlier high-dimensional layers so RO on the final layers is not very helpful. Additionally, we observed that the macro-F1 even decreased slightly on the final Adam_no_biaccorr+earlyStopping model after RO. The decision rule for Adult is clear: prioritize optimizer selection, then regularization, use RO only if compute is abundant and marginal gains are acceptable.

On the wine dataset, the hypothesis that RO would show greater benefit was only partially supported. Wine’s low dimensional continuous features and noisy labels result in many shallow local minima, this means that local search may occasionally outperform SGD. RHC and GA sometimes had improvements over SGD on certain seeds but these gains were inconsistent across runs. The Wine results confirm that it has shallow minima but show that noise restricts consistent results. These show that the effectiveness of optimization and regularization methods depends on the structure of the dataset. Sparse and imbalanced datasets seem to benefit from adaptive optimizers and early stopping but shallow and noisy surfaces have small gains from RO.

AI Use Statement

ChatGPT and Google Gemini were used to generate templates for training and testing parts of the models. Additionally, after completing the report I used Google Gemini to correct spelling and grammar errors.

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